Regression with Neural Networks - Case Study

Problem Statement

Let's consider a dataset containing measurements related to some rare gem stones, with 2 measured features and the sale price. Our final goal would be to try to predict the sale price of a new gem stone we just mined from the ground, in order to try to set a fair price in the market.

Import Data Analysis and Visualization Libraries

In [29]:
```
%pip install numpy pandas matplotlib seaborn openpyxl
%pip install ipywidgets

import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
```

```
Requirement already satisfied: numpy in /usr/local/python/3.14.2/lib/python
3.14/site-packages (2.5.4)
Requirement already satisfied: pandas in /usr/local/python/3.14.2/lib/python
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ace/.local/lib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets)
```

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ib/python3.14/site-packages (from ipython>=6.1.0->ipywidgets) (0.6.3)
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2.4)
Note: you may need to restart the kernel to use updated packages.
```

Import Dataset

In [30]:
```
df = pd.read_csv('gem_stones.csv')            # Import Dataset from .csv
df.head()                                     # Show header
```

Out[30]:

	price	feature1	feature2
0	461.527929	999.787558	999.766096
1	548.130011	998.861615	1001.042403
2	410.297162	1000.070267	998.844015
3	540.382220	999.952251	1000.440940
4	546.024553	1000.446011	1000.338531

In [31]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 3 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   price     1000 non-null   float64
 1   feature1  1000 non-null   float64
 2   feature2  1000 non-null   float64
dtypes: float64(3)
memory usage: 23.6 KB
```

In [32]:
```
df.describe()
```

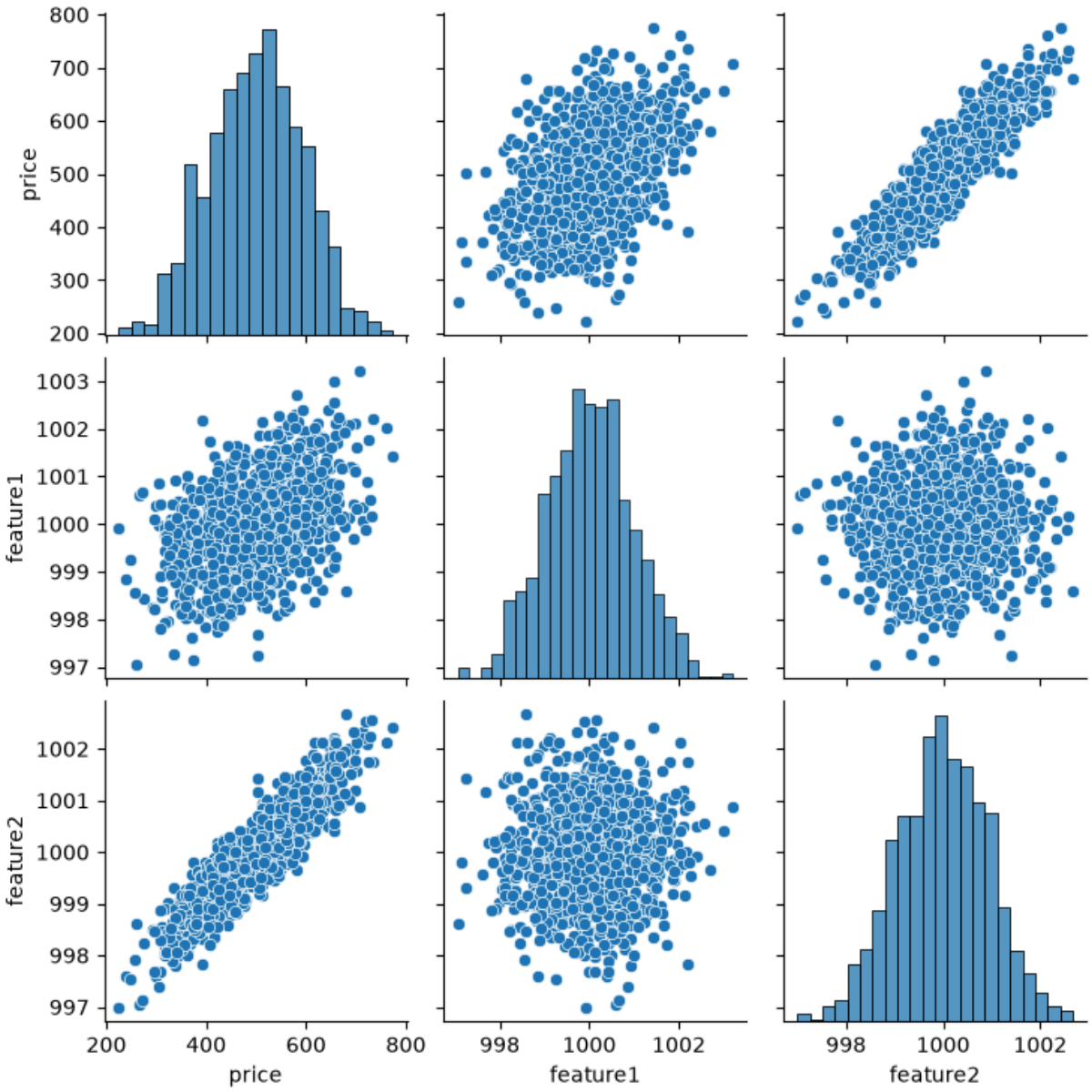
Out[32]:

	price	feature1	feature2
count	1000.000000	1000.000000	1000.000000
mean	498.673029	1000.014171	999.979847
std	93.785431	0.974018	0.948330
min	223.346793	997.058347	996.995651
25%	433.025732	999.332068	999.316106
50%	502.382117	1000.009915	1000.002243
75%	564.921588	1000.637580	1000.645380
max	774.407854	1003.207934	1002.666308

Exploratory Data Analysis

In [33]:
```
sns.pairplot(df)
```

Out[33]:
```
<seaborn.axisgrid.PairGrid at 0x74c8a737de50>
```



An approximated linear relationship is fairly evident between feature2 and price.

Test/Train Set Split

In [34]:
```
from sklearn.model_selection import train_test_split        # Import train_t

# Features (convert from DataFrame to NumPy array)
X = np.array(df[['feature1','feature2']])

# Target Variable
y = np.array(df['price'])

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, rar
print(X_train.shape, X_test.shape, y_train.shape, y_test.shape)
```

```
(700, 2) (300, 2) (700,) (300,)
```

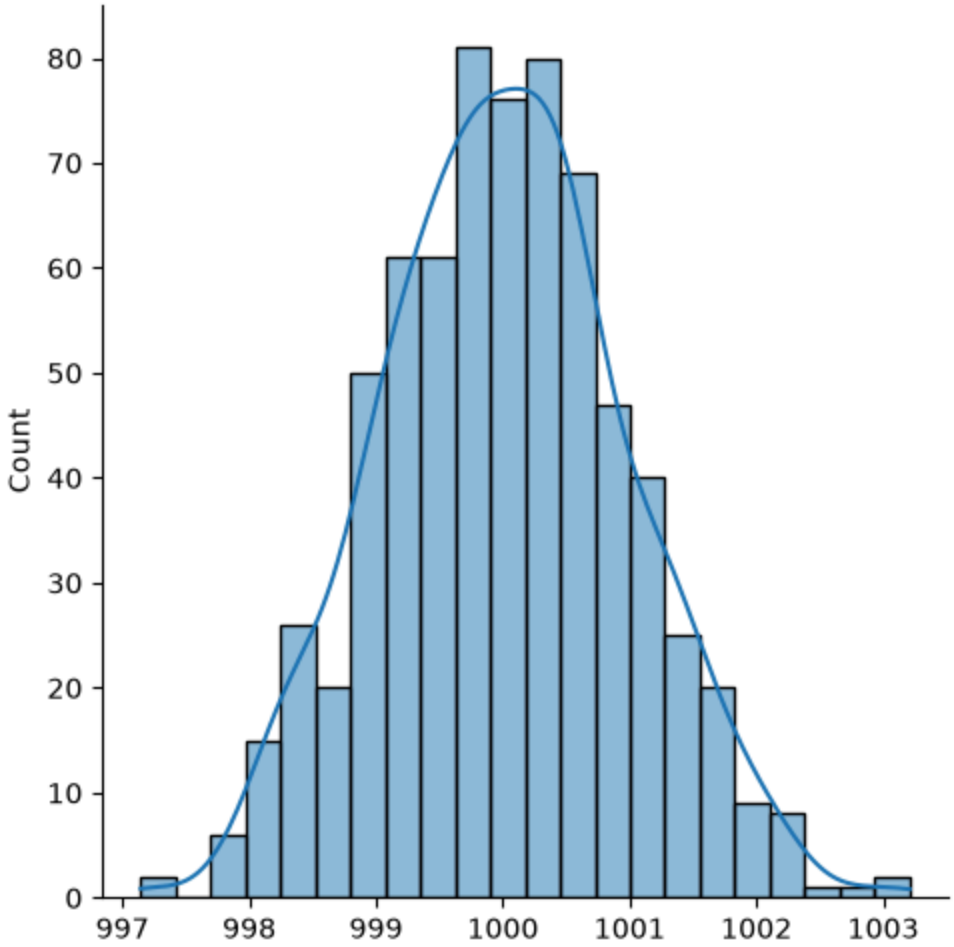
Data Pre-Processing

Use StandardScaler() to perform input centering and input normalization.

Notice that it is sufficient to standardize just input features, while it is not necessary to standardize the target variable.

In [35]:
```
sns.displot(X_train[:,0], kde = True) # distribution
```

Out[35]:
```
<seaborn.axisgrid.FacetGrid at 0x74c8a5db0910>
```



Standard scaler: $Z = \frac{X-\mu}{\sigma}$

Notice: to prevent data leakage (use information from data outside the training set) from the test set, we only fit our scaler on the training set.

In [36]:
```
from sklearn.preprocessing import StandardScaler

# high input implies small weight, so calculation problem, accuracy
# also for Gradient descent we will minimize only for un input
# output usually do not need to be scaled, because we compute gradient wrt t
```

```
# StandardScaler() object instantiation
scaler = StandardScaler()

# Fit Standard Scaler only on the training set
scaler.fit(X_train)

# Apply standardization
X_train = scaler.transform(X_train)
X_test = scaler.transform(X_test)
```

Showing importance of scaling

In [37]:
```
from sklearn.linear_model import SGDRegressor
from sklearn.metrics import mean_squared_error

# Synthetic data
np.random.seed(0)
engine_size = np.random.uniform(1, 6, 100)
weight = np.random.uniform(800, 3000, 100)
y_example = 40 - 3 * engine_size - 0.005 * weight + np.random.randn(100) * 2

# Case 1: Without normalization
X_raw = np.column_stack([engine_size, weight])
model_raw = SGDRegressor(max_iter=1000, tol=1e-3).fit(X_raw, y_example)
print("Without normalization MSE:", mean_squared_error(y_example, model_raw.

# Case 2: With normalization
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_raw)
model_scaled = SGDRegressor(max_iter=1000, tol=1e-3).fit(X_scaled, y_example
print("With normalization MSE:", mean_squared_error(y_example, model_scaled.
```

```
Without normalization MSE: 1.6290165923052177e+31
With normalization MSE: 3.3285464168765406
```

Neural Network Definition - TensorFlow & Keras

There are several ways you can import Keras from Tensorflow. We will use the method shown in the **official TF documentation**.

Creating a NN Model

In [38]:
```
%pip install tensorflow
from tensorflow.keras.models import Sequential                # Import Sequer
from tensorflow.keras.layers import Dense                     # Import Dense(
```

```
Requirement already satisfied: tensorflow in /usr/local/python/3.14.2/lib/py
thon3.14/site-packages (2.22.0rc0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/python/3.14.2/li
b/python3.14/site-packages (from tensorflow) (2.5.1)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/python/3.14.
2/lib/python3.14/site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in /usr/local/python/3.1
4.2/lib/python3.14/site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /usr/l
ocal/python/3.14.2/lib/python3.14/site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/python/3.1
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Requirement already satisfied: libclang>=13.0.0 in /usr/local/python/3.14.2/
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Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/python/3.14.
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Requirement already satisfied: packaging in /home/codespace/.local/lib/pytho
n3.14/site-packages (from tensorflow) (26.3)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in /usr/local/python/
3.14.2/lib/python3.14/site-packages (from tensorflow) (7.36.2)
Requirement already satisfied: requests<3,>=2.21.0 in /home/codespace/.loca
l/lib/python3.14/site-packages (from tensorflow) (2.34.2)
Requirement already satisfied: six>=1.12.0 in /home/codespace/.local/lib/pyt
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lib/python3.14/site-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /home/codespace/.
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Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/python/3.1
4.2/lib/python3.14/site-packages (from tensorflow) (1.85.1)
Requirement already satisfied: keras-nightly>=3.12.0.dev in /usr/local/pytho
n/3.14.2/lib/python3.14/site-packages (from tensorflow) (3.16.0.dev202610310
3)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/python/3.14.2/li
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Requirement already satisfied: h5py~=3.15.1 in /usr/local/python/3.14.2/lib/
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Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/python/
3.14.2/lib/python3.14/site-packages (from tensorflow) (0.6.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /home/codespace/.
local/lib/python3.14/site-packages (from requests<3,>=2.21.0->tensorflow)
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23)
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14/site-packages (from keras-nightly>=3.12.0.dev->tensorflow) (15.0.0)
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```

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2.0.dev->tensorflow) (0.1.2)
Note: you may need to restart the kernel to use updated packages.
```

NN Model - Adding layers one by one

Sequential model

- When to use it: A Sequential model is appropriate for a plain stack of layers where each layer has exactly one input tensor and one output tensor.
- When **not** to use it:
  - Your model has multiple inputs or multiple outputs
  - Any of your layers has multiple inputs or multiple outputs
  - You need to do layer sharing
  - You want non-linear topology (e.g. a residual connection, a multi-branch model)
- How to use it: You can create a Sequential model by passing a list of layers to the Sequential constructor

In [ ]:
```python
model = Sequential()

# Model with 3 hidden layers, wach with 4 neurons and ReLU activation functi
model.add(Dense(4, activation='relu'))
model.add(Dense(4, activation='relu'))
model.add(Dense(4, activation='relu'))
# more complx is the model (more neurons), faster is the convergence at the
# converge due to noise and complexity

# Final output node for prediction
# In the output layer we just have one neuron since the target is a single v
model.add(Dense(1))

# Optimizer determines the way in which you approach gradient descent (alter
model.compile(optimizer='rmsprop', loss='mse')
```

Optimizer: The optimizer adjusts the network's weights to minimize the loss function (e.g., Mean Squared Error in regression). It determines how the model learns from the data by controlling the step size and direction of each update.

Examples:

- SGD (Stochastic Gradient Descent): updates weights using gradients from small batches; simple but can be slow.
- Adam: combines momentum and adaptive learning rates, offering faster and more stable convergence — widely used in regression and general tasks.
- RMSprop: adapts the learning rate based on recent gradient magnitudes, effective for noisy or non-stationary data.

Choosing suitable Loss functions and Evaluation Metrics

Gradient Descent:
Gradient descent is an optimization algorithm used to minimize a loss function by iteratively updating the model's parameters in the direction of the *negative gradient*. The idea is to move step by step toward the point where the loss is smallest.

At each iteration, the weights w are updated as:

$$w \leftarrow w - \eta \nabla_w L(w)$$

where η is the learning rate (step size) and $\nabla_w L(w)$ is the gradient of the loss with respect to the weights.

In essence, gradient descent helps the neural network learn by gradually adjusting its parameters to reduce prediction error.

Keep in mind what kind of problem you are trying to solve:

```
# For a multi-class classification problem
model.compile(optimizer='rmsprop',
              loss='categorical_crossentropy',
              metrics='accuracy')

# For a binary classification problem
model.compile(optimizer='rmsprop',
              loss='binary_crossentropy',
              metrics='accuracy')

# For a mean squared error regression problem
model.compile(optimizer='rmsprop',
              loss='mse')
```

Training NN Model

In [40]:
```
model.fit(X_train, y_train, epochs=250)
```

```
Epoch 1/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/step - loss: 256779.2031
Epoch 2/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 256720.5469
Epoch 3/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 256678.3594
Epoch 4/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 256640.4062
Epoch 5/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 256602.8750
Epoch 6/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 256559.4219
Epoch 7/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 256500.4844
Epoch 8/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 256422.5781
Epoch 9/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 256328.6562
Epoch 10/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 256219.3125
Epoch 11/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 256094.3594
Epoch 12/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 255952.3906
Epoch 13/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 255791.0469
Epoch 14/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 255607.9844
Epoch 15/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 255400.7031
Epoch 16/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 255167.0156
Epoch 17/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 254903.7031
Epoch 18/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 254609.0781
Epoch 19/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 254280.3594
Epoch 20/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 253914.6562
Epoch 21/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 253507.5156
Epoch 22/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 253053.9375
Epoch 23/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 252550.2188
Epoch 24/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 251992.8438
Epoch 25/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 251378.0625
Epoch 26/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 250701.1406
Epoch 27/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 249957.8125
Epoch 28/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 249144.2500
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Epoch 29/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 248255.2188
Epoch 30/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 247286.4531
Epoch 31/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 246234.0156
Epoch 32/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 245091.9375
Epoch 33/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 243856.8438
Epoch 34/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 242524.1094
Epoch 35/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 241089.3906
Epoch 36/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 239547.0469
Epoch 37/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 237892.4531
Epoch 38/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 236122.1094
Epoch 39/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 234231.4531
Epoch 40/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 232215.3438
Epoch 41/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 230070.1250
Epoch 42/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 227792.7969
Epoch 43/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 225379.3906
Epoch 44/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 222826.2656
Epoch 45/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 220130.3281
Epoch 46/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 217287.8125
Epoch 47/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 214297.6406
Epoch 48/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 211154.9062
Epoch 49/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 207860.4844
Epoch 50/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 204414.0312
Epoch 51/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 200815.2031
Epoch 52/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 197063.7656
Epoch 53/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 193160.8438
Epoch 54/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 189108.2188
Epoch 55/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 184907.2969
Epoch 56/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 180562.2656
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Epoch 57/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 176078.6562
Epoch 58/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 171462.1875
Epoch 59/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 166720.2812
Epoch 60/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 161860.1562
Epoch 61/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 156891.9531
Epoch 62/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 151826.4844
Epoch 63/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 146677.1094
Epoch 64/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 141457.5938
Epoch 65/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 136183.5625
Epoch 66/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 130871.6797
Epoch 67/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 125541.1875
Epoch 68/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 120213.4297
Epoch 69/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 114912.5391
Epoch 70/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 109662.7578
Epoch 71/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 104487.0938
Epoch 72/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 99403.7266
Epoch 73/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 94431.4531
Epoch 74/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 89570.4609
Epoch 75/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 84817.0234
Epoch 76/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 80177.7188
Epoch 77/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 75655.0781
Epoch 78/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 71253.5703
Epoch 79/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 67005.2109
Epoch 80/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 62928.8867
Epoch 81/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 59058.8906
Epoch 82/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 55421.9023
Epoch 83/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 52006.8750
Epoch 84/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 48864.3828
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Epoch 85/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 46026.4648
Epoch 86/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 43485.9844
Epoch 87/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 41224.9531
Epoch 88/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 39221.4023
Epoch 89/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 37410.8359
Epoch 90/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 35738.6211
Epoch 91/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 34164.7539
Epoch 92/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 32657.4707
Epoch 93/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 31205.2227
Epoch 94/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 29805.3457
Epoch 95/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 28446.8066
Epoch 96/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 27117.1895
Epoch 97/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25829.7422
Epoch 98/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 24574.5488
Epoch 99/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 23348.5625
Epoch 100/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 22160.0820
Epoch 101/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 21002.4531
Epoch 102/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 19875.4609
Epoch 103/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 18786.4512
Epoch 104/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 17731.0977
Epoch 105/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 16708.3379
Epoch 106/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 15720.1182
Epoch 107/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 14765.9482
Epoch 108/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 13848.5928
Epoch 109/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 12971.2588
Epoch 110/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 12126.2715
Epoch 111/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 11314.9268
Epoch 112/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 10535.6631
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Epoch 113/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 9787.3711
Epoch 114/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 9070.9492
Epoch 115/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 8387.5166
Epoch 116/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 7738.6699
Epoch 117/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 7123.9771
Epoch 118/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 6541.3916
Epoch 119/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 5990.9390
Epoch 120/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 5463.5088
Epoch 121/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 4967.1216
Epoch 122/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 4503.3223
Epoch 123/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 4070.8633
Epoch 124/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 3666.4067
Epoch 125/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 3288.5686
Epoch 126/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 2937.3818
Epoch 127/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 2614.2161
Epoch 128/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 2316.7434
Epoch 129/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 2042.4716
Epoch 130/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1789.2391
Epoch 131/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1559.6991
Epoch 132/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1353.6824
Epoch 133/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 1170.2549
Epoch 134/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1007.3908
Epoch 135/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 864.8137
Epoch 136/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 741.5361
Epoch 137/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 636.2240
Epoch 138/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 548.8873
Epoch 139/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 477.3708
Epoch 140/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 420.2638
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Epoch 141/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 376.3348
Epoch 142/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 342.1924
Epoch 143/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 314.1681
Epoch 144/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 289.7592
Epoch 145/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 267.4261
Epoch 146/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 247.0414
Epoch 147/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 228.7824
Epoch 148/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 211.8028
Epoch 149/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 196.0479
Epoch 150/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 181.7274
Epoch 151/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 168.6708
Epoch 152/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 157.0031
Epoch 153/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 146.7005
Epoch 154/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 137.4387
Epoch 155/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 128.9068
Epoch 156/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 121.0328
Epoch 157/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 113.7963
Epoch 158/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 107.1711
Epoch 159/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 101.0909
Epoch 160/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 95.4980
Epoch 161/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 90.4769
Epoch 162/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 85.9879
Epoch 163/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 81.8707
Epoch 164/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 78.0147
Epoch 165/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 74.4248
Epoch 166/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 71.1277
Epoch 167/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 68.2186
Epoch 168/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 65.6976
```

```
Epoch 169/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 63.4214
Epoch 170/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 61.3819
Epoch 171/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 59.5360
Epoch 172/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 57.7962
Epoch 173/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 56.1619
Epoch 174/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 54.6199
Epoch 175/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 53.1458
Epoch 176/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 51.7415
Epoch 177/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 50.3976
Epoch 178/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 49.1121
Epoch 179/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 47.8932
Epoch 180/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 46.7376
Epoch 181/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 45.6424
Epoch 182/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 44.6047
Epoch 183/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 43.6218
Epoch 184/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 42.6906
Epoch 185/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 41.8087
Epoch 186/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 40.9735
Epoch 187/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 40.1825
Epoch 188/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 39.4371
Epoch 189/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 38.7351
Epoch 190/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 38.0672
Epoch 191/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 37.4324
Epoch 192/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 36.8294
Epoch 193/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 36.2581
Epoch 194/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 35.7184
Epoch 195/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 35.2040
Epoch 196/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 34.7146
```

```
Epoch 197/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 34.2241
Epoch 198/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 33.7583
Epoch 199/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 33.3140
Epoch 200/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 32.8903
Epoch 201/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 32.4864
Epoch 202/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 32.1018
Epoch 203/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 31.7355
Epoch 204/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 31.3864
Epoch 205/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 31.0519
Epoch 206/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 30.7332
Epoch 207/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 30.4298
Epoch 208/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 30.1411
Epoch 209/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 29.8667
Epoch 210/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 29.6059
Epoch 211/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 29.3584
Epoch 212/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 29.1235
Epoch 213/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 28.9008
Epoch 214/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 28.6900
Epoch 215/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 28.4904
Epoch 216/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 28.3016
Epoch 217/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 28.1233
Epoch 218/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 27.9549
Epoch 219/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 27.8082
Epoch 220/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 27.6759
Epoch 221/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 27.5479
Epoch 222/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 27.4250
Epoch 223/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 27.3068
Epoch 224/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 27.1935
```

```
Epoch 225/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 27.0847
Epoch 226/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 26.9804
Epoch 227/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 26.8804
Epoch 228/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 26.7846
Epoch 229/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 26.6928
Epoch 230/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 26.6049
Epoch 231/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 26.5208
Epoch 232/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 26.4404
Epoch 233/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 26.3636
Epoch 234/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 26.2901
Epoch 235/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 26.2200
Epoch 236/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 26.1531
Epoch 237/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 26.0892
Epoch 238/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 26.0283
Epoch 239/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25.9703
Epoch 240/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 25.9150
Epoch 241/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25.8623
Epoch 242/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25.8122
Epoch 243/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 25.7646
Epoch 244/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 25.7193
Epoch 245/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25.6761
Epoch 246/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25.6353
Epoch 247/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 25.5963
Epoch 248/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25.5590
Epoch 249/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 25.5228
Epoch 250/250
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - loss: 25.4873
```

Out[40]: `<keras.src.callbacks.history.History at 0x74c8b4c910f0>`

Evaluation

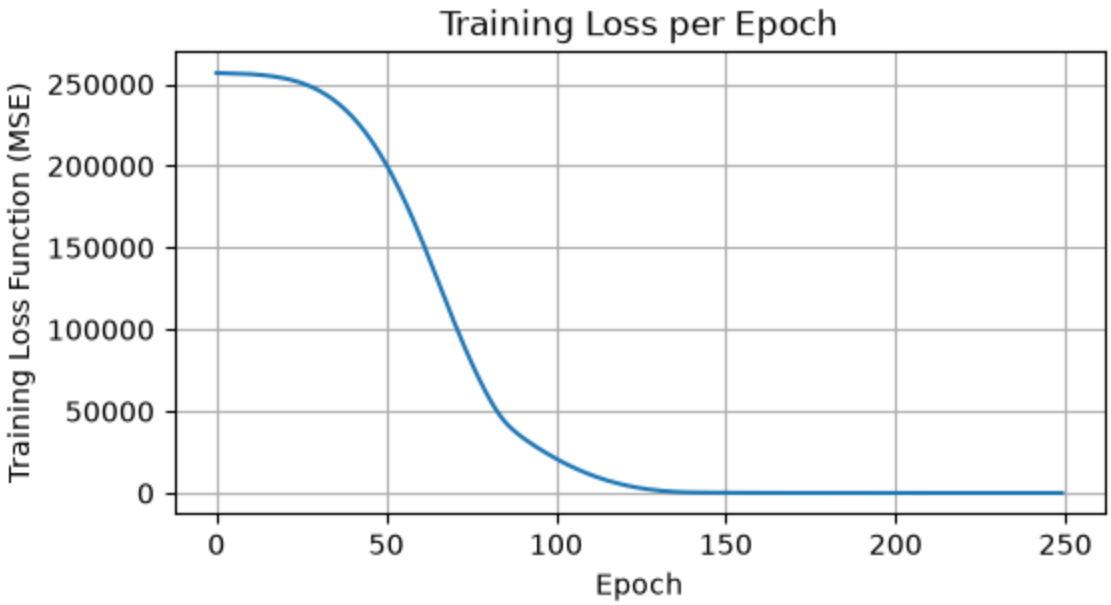
Let's evaluate our performance on our training set and our test set. We can compare these two performances to check for overfitting.

In [ ]:
```
# Check available metrics
print(model.metrics_names)

# Store performance over time
loss = model.history.history['loss']

# Plot Training Loss per Epoch
plt.figure(figsize=(6,3))
sns.lineplot(x=range(len(loss)),y=loss)
plt.title("Training Loss per Epoch")
plt.xlabel('Epoch')
plt.ylabel('Training Loss Function (MSE)')
plt.grid()
# plt.yscale('log')
```

```
['loss']
```



Loss evaluation on Training set and Test set.

These should hopefully be fairly close to each other.

In [42]:
```
# Evaluate the performance according to a metric (here, the loss function)
training_score = model.evaluate(X_train, y_train, verbose=0)
test_score = model.evaluate(X_test, y_test, verbose=0)

# Print results
print('Training score:', training_score)
print('Test score:', test_score)
```

```
Training score: 24.83881378173828
Test score: 25.70256805419922
```

Further Evaluations

In [43]:
```
# Generate the predictions and the ground truth
test_predictions = model.predict(X_test)
pred_df = pd.DataFrame(y_test, columns=['Test Y'])

# Generate a single DataFrame for both outputs
test_predictions = pd.Series(test_predictions.reshape(300,))
pred_df = pd.concat([pred_df, test_predictions],axis=1)
pred_df.columns = ['Test Y','Model Predictions']

# Display result
pred_df.head()
```

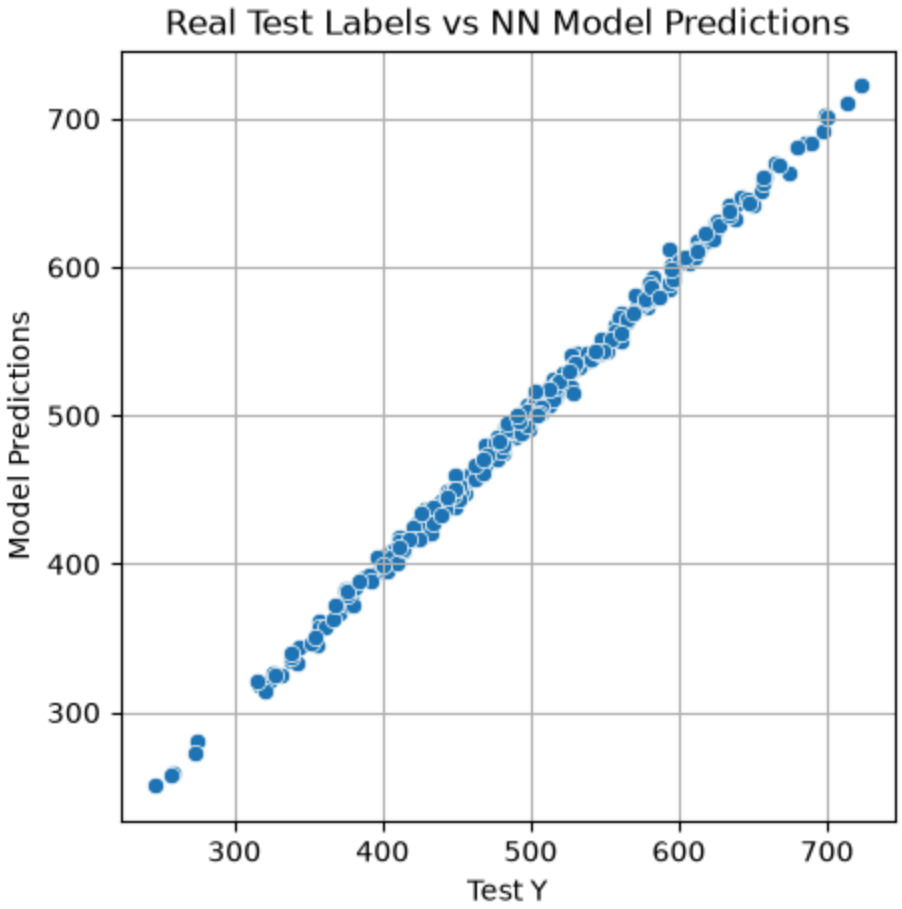
```
10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step
```

Out[43]:

	Test Y	Model Predictions
0	402.296319	405.489929
1	624.156198	624.426147
2	582.455066	592.828796
3	578.588606	573.158691
4	371.224104	367.091736

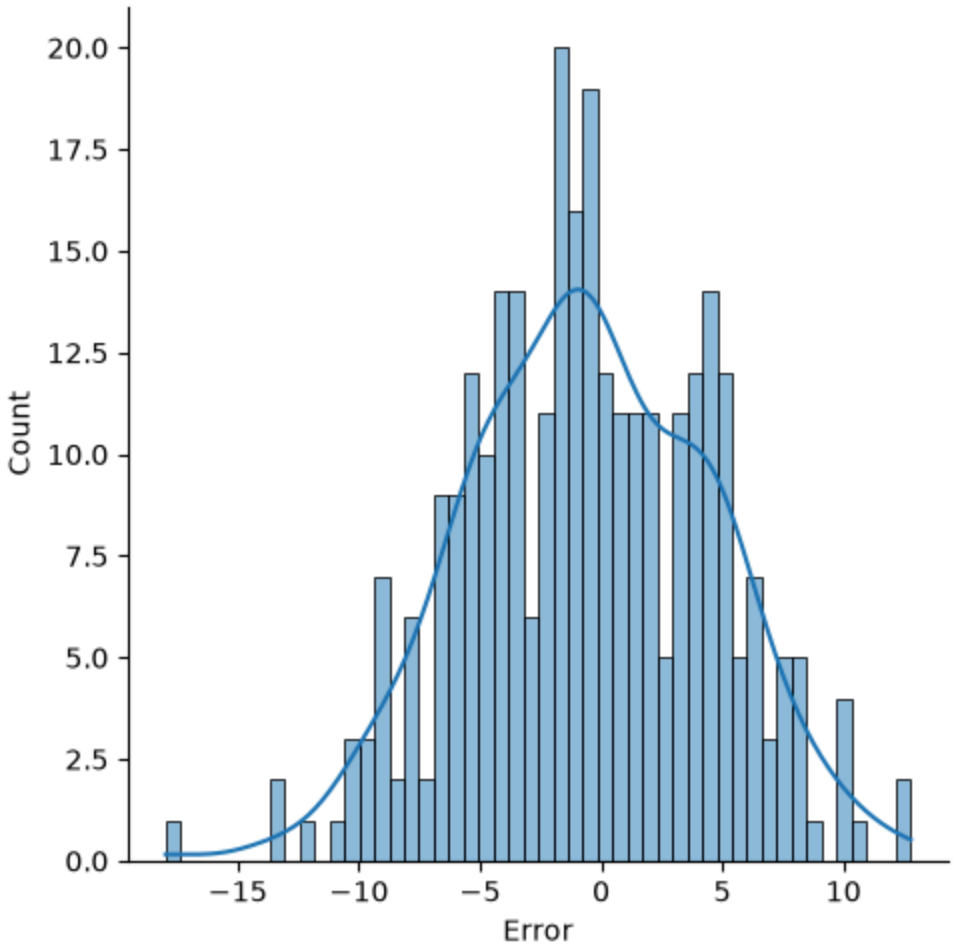
Real Test Set Labels vs NN Model Predictions

In [44]:
```
plt.figure(figsize = (5,5))
sns.scatterplot(x='Test Y', y='Model Predictions', data=pred_df)
plt.title('Real Test Labels vs NN Model Predictions')
plt.grid()
```



In [45]:
```
# Display prediction error
pred_df['Error'] = pred_df['Test Y'] - pred_df['Model Predictions']
sns.displot(pred_df['Error'], bins=50, kde=True)
```

Out[45]: <seaborn.axisgrid.FacetGrid at 0x74c8a51d4910>



In [46]:
```
# Evaluate metrics
from sklearn.metrics import mean_absolute_error, mean_squared_error
print('MAE:', mean_absolute_error(pred_df['Test Y'], pred_df['Model Predicti
print('MSE', mean_squared_error(pred_df['Test Y'], pred_df['Model Prediction
print('Test score:', test_score)
print('RMSE:', test_score**0.5)
```

```
MAE: 4.078981078965718
MSE 25.702569990906557
Test score: 25.70256805419922
RMSE: 5.0697700198528945
```

Predicting on brand new data

What if we just saw a brand new gemstone from the ground? What should we price it at? This is the **exact** same procedure as predicting on a new test data!

In [47]:
```
# [[Feature1, Feature2]]
new_gem = [[998, 1000]]

# Don't forget to standardize the new input!
new_gem = scaler.transform(new_gem)
```

```
# Predict
model.predict(new_gem)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step
```

Out[47]:
```
array([[26382.225]], dtype=float32)
```

Validation set

Useful best practice use a validation loss and monitor it because it's a proxy for overfitting

In [48]:
```
# Divide dataset into training and test
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, rar

# Divide training set into training and validation
X_train, X_val, y_train, y_val = train_test_split(X_train, y_train, test_siz

print(X_train.shape, X_val.shape, X_test.shape)
```

```
(630, 2) (70, 2) (300, 2)
```

In [49]:
```
# Standardize the data
scaler = StandardScaler()
scaler.fit(X_train)
X_train = scaler.transform(X_train)
X_val = scaler.transform(X_val)
X_test = scaler.transform(X_test)
X_train.shape, X_val.shape, X_test.shape
```

Out[49]:
```
((630, 2), (70, 2), (300, 2))
```

In [50]:
```
model = Sequential()

model.add(Dense(4, activation='relu'))
model.add(Dense(4, activation='relu'))
model.add(Dense(4 ,activation='relu'))

# Final output node for prediction
model.add(Dense(1))                          # In the output layer we just

model.compile(optimizer='rmsprop', loss='mse')
model.fit(X_train, y_train, epochs=250, validation_data=(X_val, y_val))
```

```
Epoch 1/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 1s 22ms/step - loss: 255892.7500 - val_loss: 2687
54.0312
Epoch 2/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 255680.9688 - val_loss: 26857
7.4688
Epoch 3/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 255514.2031 - val_loss: 26842
1.9688
Epoch 4/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 255366.9375 - val_loss: 26829
1.1875
Epoch 5/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 255240.4375 - val_loss: 26818
2.2812
Epoch 6/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 255130.0781 - val_loss: 26808
7.4062
Epoch 7/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 255025.1719 - val_loss: 26799
2.5625
Epoch 8/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 254920.0312 - val_loss: 26789
0.5625
Epoch 9/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 254807.2344 - val_loss: 26777
7.1562
Epoch 10/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 254683.6094 - val_loss: 26764
8.9062
Epoch 11/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 254546.3438 - val_loss: 26750
5.5000
Epoch 12/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 254394.0781 - val_loss: 26734
6.3750
Epoch 13/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 254225.9375 - val_loss: 26717
0.6562
Epoch 14/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 254040.5625 - val_loss: 26697
6.8125
Epoch 15/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 253836.6250 - val_loss: 26676
3.1250
Epoch 16/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 253612.9844 - val_loss: 26652
8.9375
Epoch 17/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 253368.2500 - val_loss: 26627
2.7188
Epoch 18/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 253100.7969 - val_loss: 26599
3.5625
Epoch 19/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 252809.2969 - val_loss: 26569
```

0.0938
Epoch 20/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 252492.4219 - val_loss: 26536
0.4375
Epoch 21/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 252148.3906 - val_loss: 26500
3.6562
Epoch 22/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 251775.0781 - val_loss: 26461
7.8438
Epoch 23/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 251371.4219 - val_loss: 26420
1.8438
Epoch 24/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 250935.5156 - val_loss: 26375
3.0625
Epoch 25/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 250466.0625 - val_loss: 26327
0.2500
Epoch 26/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 249961.1875 - val_loss: 26275
2.1250
Epoch 27/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 249419.2812 - val_loss: 26219
6.7812
Epoch 28/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 248838.6094 - val_loss: 26160
1.4844
Epoch 29/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 6ms/step - loss: 248217.6250 - val_loss: 26096
5.4844
Epoch 30/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 247554.0625 - val_loss: 26028
7.0000
Epoch 31/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 246846.3750 - val_loss: 25956
3.2344
Epoch 32/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 246092.7969 - val_loss: 25879
1.0625
Epoch 33/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 245291.3594 - val_loss: 25796
9.8281
Epoch 34/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 244439.7969 - val_loss: 25709
8.2812
Epoch 35/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 243536.2500 - val_loss: 25617
3.9219
Epoch 36/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 16ms/step - loss: 242580.0938 - val_loss: 2551
94.7500
Epoch 37/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 241568.6094 - val_loss: 25415
9.9219
Epoch 38/250

```
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 240499.6875 - val_loss: 25306
6.9219
Epoch 39/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 239371.0000 - val_loss: 25191
4.2812
Epoch 40/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 238181.5625 - val_loss: 25069
7.8281
Epoch 41/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 236929.5000 - val_loss: 24941
6.2500
Epoch 42/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 235612.5469 - val_loss: 24806
8.8906
Epoch 43/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 234227.5000 - val_loss: 24665
3.6875
Epoch 44/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 232772.2188 - val_loss: 24516
8.2500
Epoch 45/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 231244.5469 - val_loss: 24360
0.4531
Epoch 46/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 229642.2656 - val_loss: 24195
8.7500
Epoch 47/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 227964.1719 - val_loss: 24023
4.9219
Epoch 48/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 226210.0781 - val_loss: 23843
3.3438
Epoch 49/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 224379.9062 - val_loss: 23655
3.4062
Epoch 50/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 222471.7188 - val_loss: 23459
1.2031
Epoch 51/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 220487.4688 - val_loss: 23254
9.8125
Epoch 52/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 218426.2656 - val_loss: 23042
5.7969
Epoch 53/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 216286.5000 - val_loss: 22822
0.2500
Epoch 54/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 214068.0156 - val_loss: 22593
3.5625
Epoch 55/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 211771.0469 - val_loss: 22356
4.9531
Epoch 56/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 209395.5781 - val_loss: 22111
3.6875
```

```
Epoch 57/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 206941.3438 - val_loss: 21858
0.0156
Epoch 58/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 204409.0469 - val_loss: 21596
3.3594
Epoch 59/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 201799.0625 - val_loss: 21326
2.4219
Epoch 60/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 199111.8281 - val_loss: 21047
8.6875
Epoch 61/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 196348.8281 - val_loss: 20761
3.6250
Epoch 62/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 193510.0469 - val_loss: 20466
7.9531
Epoch 63/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 190597.2500 - val_loss: 20164
2.7031
Epoch 64/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 187612.6875 - val_loss: 19853
7.6406
Epoch 65/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 184558.5625 - val_loss: 19535
5.5000
Epoch 66/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 181436.5781 - val_loss: 19209
7.4062
Epoch 67/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 178250.0312 - val_loss: 18876
7.5938
Epoch 68/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 175002.0000 - val_loss: 18536
8.7500
Epoch 69/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 171696.2031 - val_loss: 18190
1.7500
Epoch 70/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 168335.8281 - val_loss: 17837
1.3125
Epoch 71/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 164924.1250 - val_loss: 17478
0.7031
Epoch 72/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 161463.5469 - val_loss: 17113
2.8750
Epoch 73/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 157957.2344 - val_loss: 16742
6.3750
Epoch 74/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 154410.0156 - val_loss: 16366
4.3438
Epoch 75/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 150823.1406 - val_loss: 15984
```

```
7.1719
Epoch 76/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 147198.6094 - val_loss: 15597
7.9219
Epoch 77/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 143537.0625 - val_loss: 15206
0.7188
Epoch 78/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 139842.0000 - val_loss: 14810
1.8125
Epoch 79/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 136121.8594 - val_loss: 14409
5.8750
Epoch 80/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 132374.2969 - val_loss: 14004
2.0625
Epoch 81/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 128599.7422 - val_loss: 13594
2.3438
Epoch 82/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 124804.0156 - val_loss: 13181
3.6562
Epoch 83/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 120999.7578 - val_loss: 12765
7.6172
Epoch 84/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 117186.5156 - val_loss: 12346
9.0000
Epoch 85/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 113369.7812 - val_loss: 11923
6.5938
Epoch 86/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 109558.3906 - val_loss: 11500
0.1016
Epoch 87/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 105759.4141 - val_loss: 11076
0.4062
Epoch 88/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 101979.2734 - val_loss: 10648
5.1406
Epoch 89/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 98221.6484 - val_loss: 10223
1.7578
Epoch 90/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 94495.4375 - val_loss: 98014.
9922
Epoch 91/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 90810.2109 - val_loss: 93823.
4375
Epoch 92/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 87161.8281 - val_loss: 89678.
1016
Epoch 93/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 83579.1328 - val_loss: 85585.
8047
Epoch 94/250
```

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20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 80061.0781 - val_loss: 81550.
3594
Epoch 95/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 76620.6250 - val_loss: 77593.
4844
Epoch 96/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 73263.4453 - val_loss: 73713.
5859
Epoch 97/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 69989.9453 - val_loss: 69940.
8203
Epoch 98/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 66804.4219 - val_loss: 66279.
0391
Epoch 99/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 63704.5703 - val_loss: 62723.
0000
Epoch 100/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 60686.3555 - val_loss: 59266.
9805
Epoch 101/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 57750.8320 - val_loss: 55930.
6953
Epoch 102/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 54894.8047 - val_loss: 52714.
1719
Epoch 103/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 52118.9141 - val_loss: 49621.
8945
Epoch 104/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 49429.4688 - val_loss: 46644.
2148
Epoch 105/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 46814.0039 - val_loss: 43751.
6445
Epoch 106/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 44246.3281 - val_loss: 40978.
3945
Epoch 107/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 41783.1797 - val_loss: 38357.
9805
Epoch 108/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 39440.1836 - val_loss: 35869.
4844
Epoch 109/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 37208.2070 - val_loss: 33514.
8984
Epoch 110/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 35086.0898 - val_loss: 31312.
7578
Epoch 111/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 33074.4961 - val_loss: 29257.
6445
Epoch 112/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 31165.4727 - val_loss: 27345.
6426
```

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Epoch 113/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 29358.5977 - val_loss: 25569.
2695
Epoch 114/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 27650.4375 - val_loss: 23894.
0938
Epoch 115/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 26032.2109 - val_loss: 22313.
9883
Epoch 116/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 24493.4219 - val_loss: 20836.
2402
Epoch 117/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 23027.9785 - val_loss: 19466.
5371
Epoch 118/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 21635.0195 - val_loss: 18185.
7598
Epoch 119/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 20306.7852 - val_loss: 16985.
0801
Epoch 120/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 19042.6484 - val_loss: 15859.
7588
Epoch 121/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 17838.2734 - val_loss: 14793.
1680
Epoch 122/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 16691.4082 - val_loss: 13787.
9883
Epoch 123/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 15595.4463 - val_loss: 12839.
5693
Epoch 124/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 14549.6250 - val_loss: 11951.
8340
Epoch 125/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 13559.4189 - val_loss: 11119.
3105
Epoch 126/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 12622.1973 - val_loss: 10337.
7471
Epoch 127/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 11736.6377 - val_loss: 9605.0
225
Epoch 128/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 10902.1846 - val_loss: 8919.7
344
Epoch 129/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 10116.8076 - val_loss: 8279.1
826
Epoch 130/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 9380.8838 - val_loss: 7685.28
03
Epoch 131/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 8694.0850 - val_loss: 7132.26
```

71
Epoch 132/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 8054.7983 - val_loss: 6619.91
89
Epoch 133/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 7459.4355 - val_loss: 6149.02
88
Epoch 134/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 7ms/step - loss: 6906.0928 - val_loss: 5717.90
43
Epoch 135/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 6394.4863 - val_loss: 5327.50
10
Epoch 136/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 5925.4204 - val_loss: 4974.07
42
Epoch 137/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 5497.5967 - val_loss: 4657.38
43
Epoch 138/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 5110.4248 - val_loss: 4375.79
54
Epoch 139/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 4761.8770 - val_loss: 4126.42
87
Epoch 140/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 4449.6226 - val_loss: 3905.09
91
Epoch 141/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 4170.9873 - val_loss: 3708.92
70
Epoch 142/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 3922.8818 - val_loss: 3533.39
79
Epoch 143/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 3701.7266 - val_loss: 3372.65
75
Epoch 144/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 3503.1250 - val_loss: 3223.20
63
Epoch 145/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 3323.8247 - val_loss: 3081.63
82
Epoch 146/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 3160.3130 - val_loss: 2947.96
19
Epoch 147/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 4ms/step - loss: 3011.1238 - val_loss: 2821.81
01
Epoch 148/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 2873.6567 - val_loss: 2701.84
28
Epoch 149/250
20/20 ━━━━━━━━━━━━━━━━━━━━ **0s** 5ms/step - loss: 2745.8887 - val_loss: 2587.79
39
Epoch 150/250

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20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 2625.9978 - val_loss: 2477.83
18
Epoch 151/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 2512.6709 - val_loss: 2370.97
92
Epoch 152/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 2405.8171 - val_loss: 2267.69
29
Epoch 153/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 2305.6558 - val_loss: 2168.61
33
Epoch 154/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 2210.4531 - val_loss: 2072.58
47
Epoch 155/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 2119.8369 - val_loss: 1980.84
19
Epoch 156/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 2033.8220 - val_loss: 1893.09
24
Epoch 157/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1951.4340 - val_loss: 1808.43
90
Epoch 158/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 1871.8516 - val_loss: 1726.60
12
Epoch 159/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1794.3899 - val_loss: 1647.95
80
Epoch 160/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 1720.4330 - val_loss: 1572.58
14
Epoch 161/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1649.4907 - val_loss: 1500.43
42
Epoch 162/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1580.3346 - val_loss: 1430.46
85
Epoch 163/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1512.8191 - val_loss: 1363.41
17
Epoch 164/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1448.4260 - val_loss: 1298.23
17
Epoch 165/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1386.8260 - val_loss: 1234.99
16
Epoch 166/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1327.4902 - val_loss: 1174.13
76
Epoch 167/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1270.2223 - val_loss: 1115.32
59
Epoch 168/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1215.2163 - val_loss: 1059.39
78
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Epoch 169/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1163.0618 - val_loss: 1005.20
09
Epoch 170/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 1113.3296 - val_loss: 952.928
5
Epoch 171/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1065.6628 - val_loss: 903.762
3
Epoch 172/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 1019.5038 - val_loss: 854.392
6
Epoch 173/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 975.1340 - val_loss: 807.0778
Epoch 174/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 932.5863 - val_loss: 761.8879
Epoch 175/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 891.8091 - val_loss: 718.5101
Epoch 176/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 852.5769 - val_loss: 677.1935
Epoch 177/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 814.8522 - val_loss: 638.1115
Epoch 178/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 778.7917 - val_loss: 600.9575
Epoch 179/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 744.3037 - val_loss: 566.0007
Epoch 180/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 711.2717 - val_loss: 532.6959
Epoch 181/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 679.5641 - val_loss: 501.1574
Epoch 182/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 649.2430 - val_loss: 470.4373
Epoch 183/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 620.0571 - val_loss: 440.8517
Epoch 184/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 591.7221 - val_loss: 412.8290
Epoch 185/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 564.2189 - val_loss: 386.1298
Epoch 186/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 537.8389 - val_loss: 361.1112
Epoch 187/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 512.9127 - val_loss: 337.7698
Epoch 188/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 488.9083 - val_loss: 315.7899
Epoch 189/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 465.7115 - val_loss: 294.9217
Epoch 190/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 443.4234 - val_loss: 274.1841
Epoch 191/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 422.1819 - val_loss: 255.1976
Epoch 192/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 401.9930 - val_loss: 237.5793
Epoch 193/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 382.5343 - val_loss: 220.8871
Epoch 194/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - loss: 363.4721 - val_loss: 204.9678
```

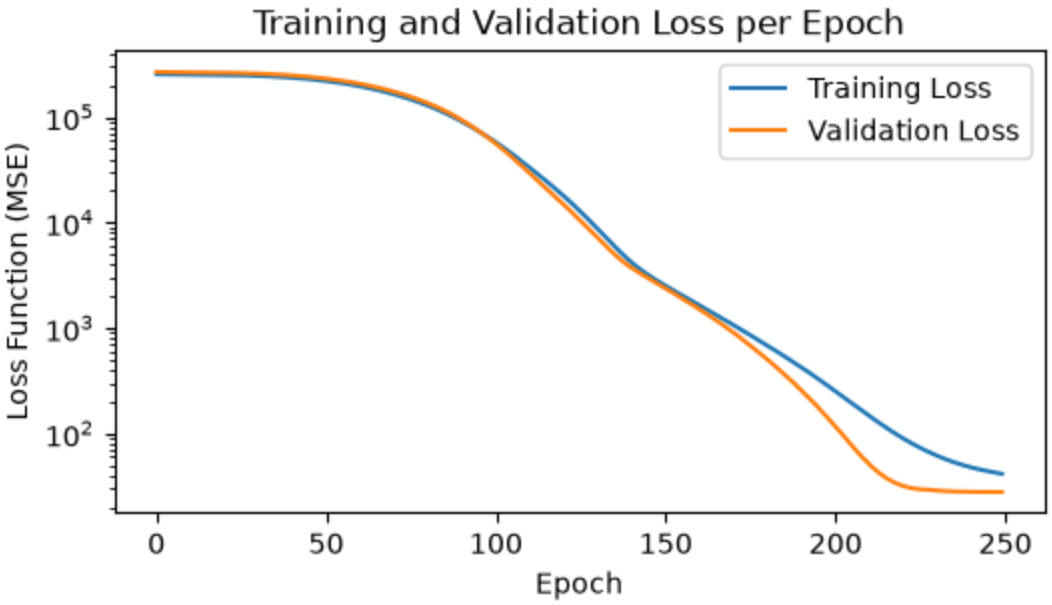
```
Epoch 195/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 345.1419 - val_loss: 189.535
6
Epoch 196/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 327.6740 - val_loss: 175.082
9
Epoch 197/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - loss: 310.7383 - val_loss: 161.507
1
Epoch 198/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 294.5553 - val_loss: 148.9022
Epoch 199/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 279.3104 - val_loss: 137.0627
Epoch 200/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 264.8403 - val_loss: 126.2496
Epoch 201/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 251.0055 - val_loss: 116.5860
Epoch 202/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 237.7974 - val_loss: 106.9604
Epoch 203/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 225.2589 - val_loss: 98.0215
Epoch 204/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 213.3156 - val_loss: 89.7973
Epoch 205/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 202.0461 - val_loss: 82.2419
Epoch 206/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 191.3629 - val_loss: 75.3685
Epoch 207/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 181.3255 - val_loss: 69.2513
Epoch 208/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 171.7932 - val_loss: 63.7609
Epoch 209/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 162.7623 - val_loss: 58.9051
Epoch 210/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 154.2039 - val_loss: 54.6637
Epoch 211/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 146.1806 - val_loss: 50.8181
Epoch 212/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 138.6539 - val_loss: 47.3425
Epoch 213/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 131.5296 - val_loss: 44.3838
Epoch 214/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 124.9023 - val_loss: 41.8176
Epoch 215/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 118.7290 - val_loss: 39.5732
Epoch 216/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 112.9862 - val_loss: 37.6400
Epoch 217/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 107.5827 - val_loss: 36.0387
Epoch 218/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 102.5403 - val_loss: 34.6354
Epoch 219/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 97.7721 - val_loss: 33.4642
Epoch 220/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 93.3534 - val_loss: 32.4918
Epoch 221/250
```

```
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 89.2695 - val_loss: 31.6881
Epoch 222/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 85.4964 - val_loss: 31.0250
Epoch 223/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 82.0262 - val_loss: 30.4931
Epoch 224/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 78.7780 - val_loss: 30.0407
Epoch 225/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 75.6953 - val_loss: 29.7327
Epoch 226/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 72.8498 - val_loss: 29.5488
Epoch 227/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 70.2222 - val_loss: 29.4384
Epoch 228/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 67.7426 - val_loss: 29.2733
Epoch 229/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 65.4248 - val_loss: 29.0695
Epoch 230/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 63.2701 - val_loss: 28.8826
Epoch 231/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 61.2646 - val_loss: 28.7185
Epoch 232/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 59.4081 - val_loss: 28.5725
Epoch 233/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 57.6707 - val_loss: 28.4397
Epoch 234/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 56.0462 - val_loss: 28.3368
Epoch 235/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 54.5307 - val_loss: 28.2355
Epoch 236/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 53.1430 - val_loss: 28.1556
Epoch 237/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 51.8729 - val_loss: 28.1036
Epoch 238/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 50.6967 - val_loss: 28.0517
Epoch 239/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 49.6033 - val_loss: 28.0088
Epoch 240/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 48.5744 - val_loss: 27.9717
Epoch 241/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 47.6034 - val_loss: 27.9443
Epoch 242/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 46.7003 - val_loss: 27.9193
Epoch 243/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 45.8639 - val_loss: 27.8909
Epoch 244/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 45.1002 - val_loss: 27.8851
Epoch 245/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 44.3953 - val_loss: 27.8734
Epoch 246/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 43.7379 - val_loss: 27.8740
Epoch 247/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 43.1128 - val_loss: 27.8669
Epoch 248/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 42.5155 - val_loss: 27.8611
Epoch 249/250
```

```
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 41.9440 - val_loss: 27.8553
Epoch 250/250
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 41.3966 - val_loss: 27.8474
```

Out[50]: <keras.src.callbacks.history.History at 0x74c8a7381f30>

In [51]:
```
# Plot Training and Validation Loss per Epoch
loss = model.history.history['loss']
val_loss = model.history.history['val_loss']
plt.figure(figsize=(6,3))
sns.lineplot(x=range(len(loss)), y=loss, label='Training Loss')
sns.lineplot(x=range(len(val_loss)), y=val_loss, label='Validation Loss')
plt.title("Training and Validation Loss per Epoch")
plt.xlabel('Epoch')
plt.ylabel('Loss Function (MSE)')
plt.legend()
plt.yscale('log')
plt.show()
```



Saving and Loading a Model

In [52]:
```
from tensorflow.keras.models import load_model

# Save model (creates a HDF5 file 'my_model.h5')
model.save('my_model.keras')

# Load model
saved_model = load_model('my_model.keras')

# Predict with loaded model
saved_model.predict(new_gem)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 52ms/step
```

Out[52]: array([[52600.574]], dtype=float32)